

Kunal Maheshwari

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SUMMARY

OPT-EAD holder and Master of Science graduate with over 4 years of experience as a Software Developer and Data Engineer, specializing in scalable microservices and distributed data pipelines. Proficient in Java, Python, Spark, SQL, AWS, and Azure, with a strong command of CI/CD automation and privacy-preserving AI applications. Seeking to leverage expertise in distributed systems, workflow automation, and SaaS product development as a Senior Backend Engineer, utilizing C++ and Go for advanced programming tasks.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime, ensuring robust workflow.
- Engineered robust ETL pipelines and automated CI/CD workflows using Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and improving monitoring and error handling, reducing issue resolution time by 20 percent.
- Spearheaded a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles for critical product features.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, maintaining a consistent track record of on-time delivery.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, contributing to core SaaS product development.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code, impacting product features.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, significantly enhancing workflow efficiency.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a culture of technical excellence.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization, ensuring reliable service delivery.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity for critical workflows.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and contributing to stable software development.
- Supported release planning and sprint execution using JIRA within Agile development teams, demonstrating effective task tracking and collaboration.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Docker, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline to process and analyze more than 8 GB of NYC taxi trip data in near real time, enabling critical data insights.
- Developed an interactive Flask web application for data exploration with dynamic filtering and visualization, enhancing user interaction with product features.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and efficient workflow management.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and improving distributed systems performance.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, for robust infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, demonstrating advanced C++ and Go programming potential.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, showcasing an understanding of Monitor and Saas principles.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, ensuring data privacy and security in distributed systems.

SKILLS

- Programming Languages: Proficient in Java, Python, SQL, C++ , and Go, with foundational knowledge in Node.js for diverse software development needs.
- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit for building scalable applications.
- Cloud Platforms: Extensive experience with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform for cloud-native development and SaaS solutions.
- Databases: Skilled in PostgreSQL, Oracle Database, IBM DB2, and RDS, with a strong understanding of data warehousing and ETL pipelines.
- Tools and Technologies: Proficient with Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for CI/CD, workflow automation, and infrastructure management.
- Concepts: Deep understanding of Distributed Systems, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- Data Analysis and Visualization: Experienced in Tableau and QuickSight for creating insightful dashboards and supporting product development.
- Soft Skills: Proven track record in problem-solving, leadership, and effective communication within cross-functional teams, adept at navigating complex litigation support scenarios.